

Lab 10

```
setwd("C:\\Users\\it24103513\\Desktop\\IT24103513")

#Exercise
#Q1

#Null hypothesis will be the customers choose the four snack types (A, B, C, D) with equal probability of 0.25.
#Alternative hypothesis will be the customers do not choose the four snack types (A, B, C, D) with equal probability.

observed <- c(120,95,85,100)
prob <- c(0.25,0.25,0.25,0.25)

chisq.test(x=observed, p=prob)

#Consider 5% level of significance for the test.

#Rejection Region: If the p value for the test is less than 0.05,
#reject the null hypothesis at 5% level of significance.

#Conclusion: Since the p value(0.08966) is greater than 0.05,do not reject null hypothesis at 5%
#level of significance. therefore we can conclude that there is a significant association between
#snack and type
```

Console Terminal × Background Jobs ×

R 4.2.2 · C:/Users/it24103513/Desktop/IT24103513/ ↗

```
> observed <- c(120,95,85,100)
> prob <- c(0.25,0.25,0.25,0.25)
> chisq.test(x=observed, p=prob)
```

Chi-squared test for given probabilities

data: observed
X-squared = 6.5, df = 3, p-value = 0.08966

```
> |
```